Statistics Hybrid Practice Set – S7 (L3)

Topics

- Population
- Sample
- Mean
- Median
- Mode
- Range
- Population Variance
- **Population Standard Deviation (New)** ⭐

Instructions

- Solve on **paper first**.
- Then verify in **Excel**.
- Write the Excel formula for every calculation.
- No calculator.

Q1. Identify Population and Sample

A university has **10,500 students**. A survey is conducted on **350 students**.

Paper Task

- Population = ?
- Sample = ?

Excel Task

Total Students	Surveyed Students
10500	350

(No formula required.)

Q2. Calculate the Mean

Value
20
24
28
32
36

Paper Task

- Calculate the Mean.

Excel Task

- Calculate the Mean.
- Write the Excel formula.

Q3. Find the Median

Value
45
21
37
29
33

Paper Task

- Find the Median.

Excel Task

- Calculate the Median.
- Write the Excel formula.

Q4. Find the Mode

Value
10
12
12
14
16
18

Paper Task

- Find the Mode.

Excel Task

- Calculate the Mode.
- Write the Excel formula.

Q5. Calculate the Range

Value
25
42
31
19
38

Paper Task

- Calculate the Range.

Excel Task

- Calculate the Range.
- Write the Excel formula.

⭐ **New Topic – Population Standard Deviation**

Relationship: Standard Deviation = √Variance

Q6. Calculate the Population Standard Deviation

Value
2
4
6
8
10

Paper Task

1. Find the Mean.
2. Find the Population Variance.
3. Calculate the Population Standard Deviation.

Excel Task

- Calculate the Population Standard Deviation.
- Write the Excel formula.

Q7. Calculate the Population Standard Deviation

Value
7
7
7
7
7

Paper Task

- Calculate the Population Standard Deviation.

Excel Task

- Calculate the Population Standard Deviation.
- Write the Excel formula.

Q8. Interpretation

Machine A

Value
49
50
51
50
50

Machine B

Value
30
40
50
60
70

Paper Task

1. Which machine has the **higher standard deviation**?
2. Which machine is **more consistent**?
3. Explain your answer in one sentence.

Excel Task

Calculate the Population Standard Deviation for both machines.

Q9. Real Data Analyst Scenario

Two customer support teams have the **same average response time**.

- **Team A:** Response times are very similar every day.
- **Team B:** Response times vary significantly every day.

Which statistical measure best identifies the team with **more consistent performance**?

A. Mean

B. Median

C. Standard Deviation

D. Mode

Explain your answer in one sentence.

Q10. Excel Formula Challenge

Write the Excel formulas for:

1. Mean
2. Median
3. Mode
4. Range
5. Population Variance
6. Population Standard Deviation

Use the range **B2:B6**.